

Overview:

- 1) Exploratory Data Analysis (EDA)**
- 2) Outlier Screening and Handling Procedures**
- 3) Variable Selection Procedures**
- 4) Goodness of Fit Testing and Model Assumptions**
- 5) Model Training and Evaluation**
- 6) Statistical Analysis and Hypothesis Testing**
- 7) Cross-validation and Model Selection Procedures**

1) Exploratory Data Analysis: ¶

Import libraries:

```
In [1]: library(readr)
library(dplyr)
library(tidyr)
library(tidyverse)
library(janitor)
library(corrplot)
library(ggplot2)
library(hexbin)
library(car)
library(caret)
set.seed(111)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ forcats	1.0.1	✓ purrr	1.0.4
✓ ggplot2	3.5.2	✓ stringr	1.5.1
✓ lubridate	1.9.4	✓ tibble	3.2.1

— Conflicts — tidyverse_conflicts() —

* dplyr::filter() masks stats::filter()

* dplyr::lag() masks stats::lag()

i Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become errors

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

corrplot 0.95 loaded

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:purrr':

some

The following object is masked from 'package:dplyr':

recode

Loading required package: lattice

Attaching package: 'caret'

The following object is masked from 'package:purrr':

lift

Load Data:

```
In [2]: df <- read_csv( "~/Documents/ISYE_6414_Final_Project-Adrian/1_data_loading_cleaning/merged_data/merged_stat
names(df)
```

Rows: 229188 Columns: 52
— Column specification —
Delimiter: ","
chr (11): OPEID6.x, INSTNM.x, CIPCODE, CIPDESC, CREDDISC, OPEID, OPEID6.y, I...
dbl (40): UNITID, CREDLEV, CONTROL, REGION, ADM_RATE, ADM_RATE_ALL, SAT_AVG,...
lgl (1): LOCALE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

'UNITID' · 'OPEID6.x' · 'INSTNM.x' · 'CIPCODE' · 'CIPDESC' · 'CREDLEV' · 'CREDDISC' · 'OPEID' · 'OPEID6.y' · 'INSTNM.y' ·
'CITY' · 'STABBR' · 'ZIP' · 'CONTROL' · 'REGION' · 'LOCALE' · 'ADM_RATE' · 'ADM_RATE_ALL' · 'SAT_AVG' · 'ACTCMMID' ·
'TUITFTE' · 'INEXPTE' · 'TUITIONFEE_IN' · 'TUITIONFEE_OUT' · 'ROOMBOARD_OFF' · 'OTHEREXPENSE_OFF' ·
'BOOKSUPPLY' · 'UGDS' · 'UGDS_MEN' · 'UGDS_WOMEN' · 'UGDS_WHITE' · 'UGDS_BLACK' · 'UGDS_HISP' · 'UGDS_ASIAN' ·
'UGDS_AIAN' · 'UGDS_NHPI' · 'UGDS_2MOR' · 'UGDS_NRA' · 'UGDS_UNKN' · 'avg_hourly_earnings' · 'avg_weekly_earnings' ·
'avg_weekly_hours' · 'Debt_24' · 'Borrowers_24' · 'Debt_25-34' · 'Borrowers_25-34' · 'Debt_35-49' · 'Borrowers_35-49' ·
'Debt_50-61' · 'Borrowers_50-61' · 'Debt_62_Plus' · 'Borrowers_62_Plus'

Create Response Variable:

```
In [3]: # aggregate debt and income
df <- df %>%
mutate(Total_Debt = Borrowers_24 + `Debt_25-34` + `Debt_35-49` + `Debt_50-61` + Debt_62_Plus)%>%
mutate(Annual_Income = avg_weekly_earnings * 52)%>% # multiply by weeks in year
mutate(Debt_to_Income_Ratio = Total_Debt / Annual_Income ) # create debt to income ratio
df
```

A tibble: 229188 × 55

UNITID	OPEID6.x	INSTNM.x	CIPCODE	CIPDESC	CREDLEV	CREDDISC	OPEID	OPEID6.y	INSTNM.y	...	Borrowers
<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>	<chr>	<chr>	...	<dbl>
100654	001002	Alabama A & M University	0100	Agriculture, General.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21
100654	001002	Alabama A & M University	0101	Agricultural Business and Management.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21
100654	001002	Alabama A & M University	0109	Animal Sciences.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21
100654	001002	Alabama A & M University	0110	Food Science and Technology.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21

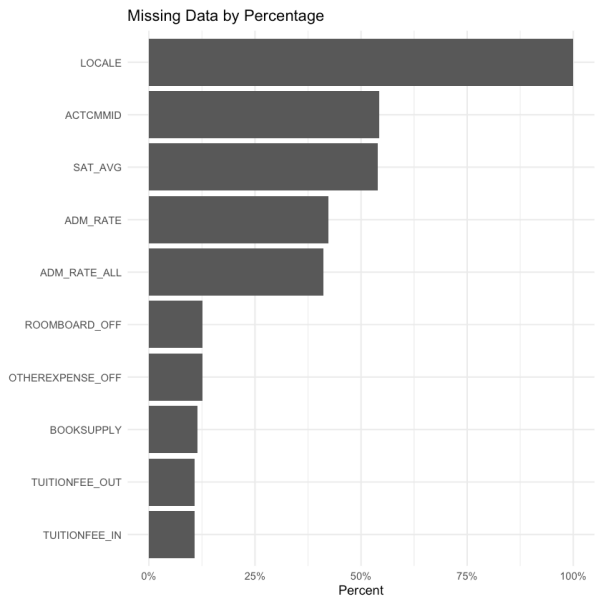
Check for Missing Data:

```
In [4]: missing_data <- df %>%
  summarise(across(everything(), ~ mean(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "column_name", values_to = "percentage_missing") %>%
  arrange(desc(percentge_missing))
  head(missing_data, 10)

# focus on columns missing more that 10% of data
missing_data %>%
  filter(percentge_missing > 0.1) %>%
  ggplot(aes(x = percentge_missing, y = fct_reorder(column_name, percentge_missing))) +
  geom_col() +
  scale_x_continuous(labels = scales::percent) +
  labs(title = "Missing Data by Percentage",
       x = "Percent", y = NULL) + theme_minimal()
```

A tibble: 10 × 2

column_name	percentage_missing
<chr>	<dbl>
LOCALE	1.0000000
ACTCMMID	0.5424586
SAT_AVG	0.5389549
ADM_RATE	0.4232421
ADM_RATE_ALL	0.4120373
ROOMBOARD_OFF	0.1257439
OTHEREXPENSE_OFF	0.1257439
BOOKSUPPLY	0.1139021
TUITIONFEE_IN	0.1081950
TUITIONFEE_OUT	0.1081950



The columns LOCALE, ACTCMMID, SAT_AVG, ADM_RATE, and ADM_RATE_ALL have high rates of missing data. Columns LOCALE, ACTCMMID, and SAT_AVG all have more than 50% of values missing. ADM_RATE and ADM_RATE_ALL are missing more than 30% of their values. Removing data with high rates is a precaution to minimize the risk of bias.

Drop Columns with High Percentage of Missing Values:

```
In [5]: df <- df [, !names(df ) %in% c("LOCALE", "ACTCMMID", "SAT_AVG", "ADM_RATE", "ADM_RATE_ALL")]
df
```

A tibble: 229188 × 50

UNITID	OPEID6.x	INSTNM.x	CIPCODE	CIPDESC	CREDLEV	CREDESC	OPEID	OPEID6.y	INSTNM.y	...	Borrowers
<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>	<chr>	<chr>	...	<dbl>
100654	001002	Alabama A & M University	0100	Agriculture, General.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21
100654	001002	Alabama A & M University	0101	Agricultural Business and Management.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21
100654	001002	Alabama A & M University	0109	Animal Sciences.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21
100654	001002	Alabama A & M University	0110	Food Science and Technology.	3	Bachelor's Degree	00100200	001002	Alabama A & M University	...	21

Check for Missing Data in Reduced df :

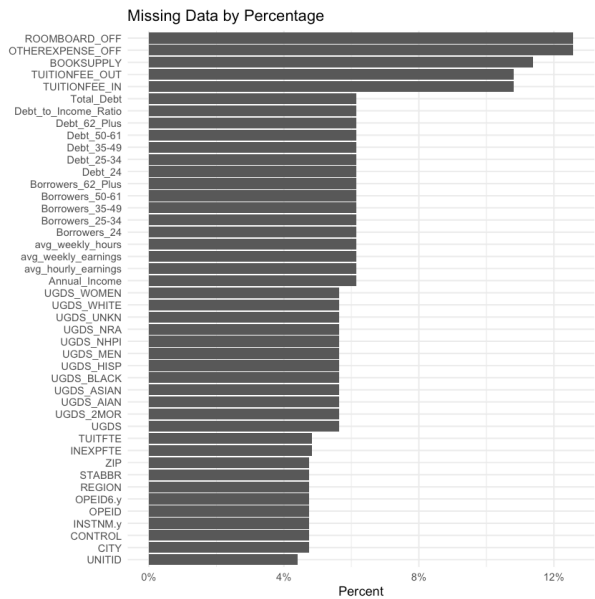
```
In [6]: missing_tbl <- df %>%
  summarise(across(everything(), ~ mean(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "column_name", values_to = "percentage_missing") %>%
  arrange(desc(percentge_missing))

head(missing_tbl, 10)

missing_tbl %>%
  filter(percentge_missing > 0.01) %>%
  ggplot(aes(x = percentge_missing, y = fct_reorder(column_name, percentge_missing))) +
  geom_col() +
  scale_x_continuous(labels = scales::percent) +
  labs(title = "Missing Data by Percentage",
       x = "Percent", y = NULL) + theme_minimal()
```

A tibble: 10 × 2

column_name	percentage_missing
<chr>	<dbl>
ROOMBOARD_OFF	0.12574393
OTHEREXPENSE_OFF	0.12574393
BOOKSUPPLY	0.11390212
TUITIONFEE_IN	0.10819502
TUITIONFEE_OUT	0.10819502
avg_hourly_earnings	0.06135574
avg_weekly_earnings	0.06135574
avg_weekly_hours	0.06135574
Debt_24	0.06135574
Borrowers_24	0.06135574



Now that LOCALE, ACTCMMID, SAT_AVG, ADM_RATE, and ADM_RATE_ALL have been removed from the df, the columns with the highest rates of missing values are ROOMBOARD_OFF, OTHEREXPENSE_OFF, BOOKSUPPLY, TUITIONFEE_IN, and TUITIONFEE_OUT, each with about 10% missing values. The other columns are missing 4% to 6% of values. This low proportion of missing data does not pose a significant risk of bias.

Inspect Data:

```
In [7]: names(df)
        glimpse(df)
```

```
'UNITID' · 'OPEID6.x' · 'INSTNM.x' · 'CIPCODE' · 'CIPDESC' · 'CREDLEV' · 'CREDESC' · 'OPEID' · 'OPEID6.y' · 'INSTNM.y' ·
'CITY' · 'STABBR' · 'ZIP' · 'CONTROL' · 'REGION' · 'TUITFTE' · 'INEXPFTE' · 'TUITIONFEE_IN' · 'TUITIONFEE_OUT' ·
'ROOMBOARD_OFF' · 'OTHEREXPENSE_OFF' · 'BOOKSUPPLY' · 'UGDS' · 'UGDS_MEN' · 'UGDS_WOMEN' · 'UGDS_WHITE' ·
'UGDS_BLACK' · 'UGDS_HISP' · 'UGDS_ASIAN' · 'UGDS_AIAN' · 'UGDS_NHPI' · 'UGDS_2MOR' · 'UGDS_NRA' · 'UGDS_UNKN' ·
'avg_hourly_earnings' · 'avg_weekly_earnings' · 'avg_weekly_hours' · 'Debt_24' · 'Borrowers_24' · 'Debt_25-34' ·
'Borrowers_25-34' · 'Debt_35-49' · 'Borrowers_35-49' · 'Debt_50-61' · 'Borrowers_50-61' · 'Debt_62_Plus' · 'Borrowers_62_Plus' ·
'Total_Debt' · 'Annual_Income' · 'Debt_to_Income_Ratio'
```

Rows: 229,188

Columns: 50

```
$ UNITID          <dbl> 100654, 100654, 100654, 100654, 100654, 100654, 1...
$ OPEID6.x       <chr> "001002", "001002", "001002", "001002", "001002", "...
$ INSTNM.x       <chr> "Alabama A & M University", "Alabama A & M Univer...
$ CIPCODE        <chr> "0100", "0101", "0109", "0110", "0110", "0110", "...
$ CIPDESC        <chr> "Agriculture, General.", "Agricultural Business a...
$ CREDLEV        <dbl> 3, 3, 3, 3, 5, 6, 3, 3, 5, 6, 3, 3, 5, 3, 3...
$ CREDESC        <chr> "Bachelor's Degree", "Bachelor's Degree", "Bachel...
$ OPEID          <chr> "00100200", "00100200", "00100200", "00100200", "...
$ OPEID6.y       <chr> "001002", "001002", "001002", "001002", "001002", "...
$ INSTNM.y       <chr> "Alabama A & M University", "Alabama A & M Univer...
$ CITY           <chr> "Normal", "Normal", "Normal", "Normal", "Normal", "...
$ STABBR         <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", "...
$ ZIP            <chr> "35762", "35762", "35762", "35762", "35762", "357...
$ CONTROL        <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ REGION         <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5...
$ TUITFTE        <dbl> 8101, 8101, 8101, 8101, 8101, 8101, 8101, 8101, 8...
$ INEXPFTE       <dbl> 4836, 4836, 4836, 4836, 4836, 4836, 4836, 4836, 4...
$ TUITIONFEE_IN  <dbl> 10024, 10024, 10024, 10024, 10024, 10024, 10024, ...
$ TUITIONFEE_OUT <dbl> 18634, 18634, 18634, 18634, 18634, 18634, 18634, ...
$ ROOMBOARD_OFF  <dbl> 9240, 9240, 9240, 9240, 9240, 9240, 9240, 9240, 9...
$ OTHEREXPENSE_OFF <dbl> 3090, 3090, 3090, 3090, 3090, 3090, 3090, 3090, 3...
$ BOOKSUPPLY     <dbl> 1600, 1600, 1600, 1600, 1600, 1600, 1600, 1600, 1...
$ UGDS           <dbl> 5090, 5090, 5090, 5090, 5090, 5090, 5090, 5090, 5...
$ UGDS_MEN       <dbl> 0.3978, 0.3978, 0.3978, 0.3978, 0.3978, 0.3978, 0.3978, 0...
$ UGDS_WOMEN     <dbl> 0.6022, 0.6022, 0.6022, 0.6022, 0.6022, 0.6022, 0.6022, 0...
$ UGDS_WHITE     <dbl> 0.0159, 0.0159, 0.0159, 0.0159, 0.0159, 0.0159, 0.0159, 0...
$ UGDS_BLACK     <dbl> 0.9022, 0.9022, 0.9022, 0.9022, 0.9022, 0.9022, 0.9022, 0...
$ UGDS_HISP      <dbl> 0.0116, 0.0116, 0.0116, 0.0116, 0.0116, 0.0116, 0.0116, 0...
$ UGDS_ASIAN     <dbl> 0.0012, 0.0012, 0.0012, 0.0012, 0.0012, 0.0012, 0.0012, 0...
$ UGDS_AIAN      <dbl> 0.0028, 0.0028, 0.0028, 0.0028, 0.0028, 0.0028, 0.0028, 0...
$ UGDS_NHPI      <dbl> 8e-04, 8e-04, 8e-04, 8e-04, 8e-04, 8e-04, 8e-04, ...
$ UGDS_2MOR      <dbl> 0.0143, 0.0143, 0.0143, 0.0143, 0.0143, 0.0143, 0.0143, 0...
$ UGDS_NRA       <dbl> 0.0073, 0.0073, 0.0073, 0.0073, 0.0073, 0.0073, 0.0073, 0...
$ UGDS_UNKN      <dbl> 0.044, 0.044, 0.044, 0.044, 0.044, 0.044, 0.044, ...
$ avg_hourly_earnings <dbl> 31.48, 31.48, 31.48, 31.48, 31.48, 31.48, 31.48, ...
$ avg_weekly_earnings <dbl> 1117.54, 1117.54, 1117.54, 1117.54, 1117.54, 1117.54, 1117...
$ avg_weekly_hours  <dbl> 35.5, 35.5, 35.5, 35.5, 35.5, 35.5, 35.5, 35.5, ...
$ Debt_24          <dbl> 1.71e+09, 1.71e+09, 1.71e+09, 1.71e+09, 1.71e+09, ...
$ Borrowers_24     <dbl> 111.4, 111.4, 111.4, 111.4, 111.4, 111.4, 111.4, ...
$ `Debt_25-34`     <dbl> 7.1e+09, 7.1e+09, 7.1e+09, 7.1e+09, 7.1e+09, 7.1e+09, 7.1e...
$ `Borrowers_25-34` <dbl> 204.6, 204.6, 204.6, 204.6, 204.6, 204.6, 204.6, ...
$ `Debt_35-49`     <dbl> 9.5e+09, 9.5e+09, 9.5e+09, 9.5e+09, 9.5e+09, 9.5e...
$ `Borrowers_35-49` <dbl> 192.6, 192.6, 192.6, 192.6, 192.6, 192.6, 192.6, ...
$ `Debt_50-61`     <dbl> 3.76e+09, 3.76e+09, 3.76e+09, 3.76e+09, 3.76e+09, ...
$ `Borrowers_50-61` <dbl> 83.3, 83.3, 83.3, 83.3, 83.3, 83.3, 83.3, 83.3, ...
$ Debt_62_Plus     <dbl> 1.05e+09, 1.05e+09, 1.05e+09, 1.05e+09, 1.05e+09, ...
$ Borrowers_62_Plus <dbl> 26.3, 26.3, 26.3, 26.3, 26.3, 26.3, 26.3, 26.3, 2...
$ Total_Debt       <dbl> 2.141e+10, 2.141e+10, 2.141e+10, 2.141e+10, 2.141e+10, ...
$ Annual_Income    <dbl> 58112.08, 58112.08, 58112.08, 58112.08, 58112.08, ...
$ Debt_to_Income_Ratio <dbl> 368426, 368426, 368426, 368426, 368426, 368426, 3...
```

Focusing on Numeric Columns:

```
In [8]: eda_df <- df %>%
select(where(is.numeric))
head(eda_df)
summary(eda_df)
```

A tibble: 6 × 39

UNITID	CREDLEV	CONTROL	REGION	TUITFTE	INEXPFTE	TUITIONFEE_IN	TUITIONFEE_OUT	ROOMBOARD_OFF	OTHEREXPENSE_OFF	...	B
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	...	
100654	3	1	5	8101	4836	10024	18634	9240	3090	...	
100654	3	1	5	8101	4836	10024	18634	9240	3090	...	
100654	3	1	5	8101	4836	10024	18634	9240	3090	...	
100654	3	1	5	8101	4836	10024	18634	9240	3090	...	
100654	5	1	5	8101	4836	10024	18634	9240	3090	...	
100654	6	1	5	8101	4836	10024	18634	9240	3090	...	

UNITID	CREDLEV	CONTROL	REGION
Min. :100654	Min. :1.00	Min. :1.000	Min. :0.000
1st Qu.:149231	1st Qu.:2.00	1st Qu.:1.000	1st Qu.:3.000
Median :187532	Median :3.00	Median :1.000	Median :5.000
Mean :202262	Mean :3.27	Mean :1.443	Mean :4.496
3rd Qu.:220978	3rd Qu.:5.00	3rd Qu.:2.000	3rd Qu.:6.000
Max. :497338	Max. :8.00	Max. :3.000	Max. :9.000
NA's :10109		NA's :10871	NA's :10871
TUITFTE	INEXPTE	TUITIONFEE_IN	TUITIONFEE_OUT
Min. : 0	Min. : 0	Min. : 480	Min. : 480
1st Qu.: 3845	1st Qu.: 5901	1st Qu.: 5206	1st Qu.:10104
Median : 9289	Median : 8496	Median :10198	Median :20110
Mean : 11026	Mean : 11417	Mean :17147	Mean :23437
3rd Qu.: 15254	3rd Qu.: 12300	3rd Qu.:26874	3rd Qu.:33386
Max. :795335	Max. :586058	Max. :61671	Max. :61671
NA's :11067	NA's :11067	NA's :24797	NA's :24797
ROOMBOARD_OFF	OTHEREXPENSE_OFF	BOOKSUPPLY	UGDS
Min. : 820	Min. : 210	Min. : 0	Min. : 0
1st Qu.: 8076	1st Qu.: 2800	1st Qu.:1000	1st Qu.: 1506
Median :10300	Median : 3830	Median :1200	Median : 4231
Mean :10696	Mean : 4010	Mean :1266	Mean : 8632
3rd Qu.:12876	3rd Qu.: 4960	3rd Qu.:1470	3rd Qu.: 11392
Max. :29400	Max. :21874	Max. :8144	Max. :109154
NA's :28819	NA's :28819	NA's :26105	NA's :12909
UGDS_MEN	UGDS_WOMEN	UGDS_WHITE	UGDS_BLACK
Min. :0.000	Min. :0.000	Min. :0.000	Min. :0.000
1st Qu.:0.364	1st Qu.:0.528	1st Qu.:0.363	1st Qu.:0.039
Median :0.416	Median :0.584	Median :0.561	Median :0.074
Mean :0.413	Mean :0.586	Mean :0.524	Mean :0.125
3rd Qu.:0.471	3rd Qu.:0.636	3rd Qu.:0.713	3rd Qu.:0.147
Max. :1.000	Max. :1.000	Max. :1.000	Max. :1.000
NA's :12909	NA's :12909	NA's :12909	NA's :12909
UGDS_HISP	UGDS_ASIAN	UGDS_AIAN	UGDS_NHPI
Min. :0.000	Min. :0.000	Min. :0.000	Min. :0.000
1st Qu.:0.057	1st Qu.:0.013	1st Qu.:0.001	1st Qu.:0.000
Median :0.106	Median :0.028	Median :0.003	Median :0.001
Mean :0.175	Mean :0.056	Mean :0.011	Mean :0.004
3rd Qu.:0.222	3rd Qu.:0.067	3rd Qu.:0.006	3rd Qu.:0.002
Max. :1.000	Max. :1.000	Max. :1.000	Max. :1.000
NA's :12909	NA's :12909	NA's :12909	NA's :12909
UGDS_2MOR	UGDS_NRA	UGDS_UNKN	avg_hourly_earnings
Min. :0.000	Min. :0.000	Min. :0.000	Min. :28.10
1st Qu.:0.025	1st Qu.:0.005	1st Qu.:0.009	1st Qu.:32.52
Median :0.037	Median :0.015	Median :0.020	Median :34.56
Mean :0.040	Mean :0.032	Mean :0.035	Mean :35.34
3rd Qu.:0.048	3rd Qu.:0.040	3rd Qu.:0.038	3rd Qu.:38.96
Max. :0.922	Max. :1.000	Max. :1.000	Max. :54.23
NA's :12909	NA's :12909	NA's :12909	NA's :14062
avg_weekly_earnings	avg_weekly_hours	Debt_24	Borrowers_24
Min. : 984.9	Min. :32.50	Min. :9.000e+07	Min. : 8.3
1st Qu.:1118.7	1st Qu.:33.30	1st Qu.:1.640e+09	1st Qu.:115.1
Median :1185.4	Median :34.00	Median :3.790e+09	Median :235.1
Mean :1202.1	Mean :34.06	Mean :4.121e+09	Mean :270.4
3rd Qu.:1277.9	3rd Qu.:34.40	3rd Qu.:6.740e+09	3rd Qu.:399.3
Max. :1936.0	Max. :36.30	Max. :8.670e+09	Max. :624.6
NA's :14062	NA's :14062	NA's :14062	NA's :14062
Debt_25-34	Borrowers_25-34	Debt_35-49	Borrowers_35-49
Min. :4.700e+08	Min. : 17.9	Min. :7.200e+08	Min. : 17.9
1st Qu.:7.240e+09	1st Qu.: 215.0	1st Qu.:8.580e+09	1st Qu.: 191.5
Median :1.497e+10	Median : 422.6	Median :1.552e+10	Median : 304.4
Mean :1.868e+10	Mean : 543.2	Mean :2.010e+10	Mean : 428.5
3rd Qu.:3.200e+10	3rd Qu.: 877.6	3rd Qu.:2.901e+10	3rd Qu.: 591.5
Max. :4.999e+10	Max. :1456.4	Max. :5.244e+10	Max. :1086.8
NA's :14062	NA's :14062	NA's :14062	NA's :14062
Debt_50-61	Borrowers_50-61	Debt_62_Plus	Borrowers_62_Plus
Min. :2.700e+08	Min. : 6.7	Min. :8.000e+07	Min. : 2.00
1st Qu.:3.300e+09	1st Qu.: 78.6	1st Qu.:1.020e+09	1st Qu.: 24.00
Median :7.550e+09	Median :147.9	Median :2.130e+09	Median : 48.20
Mean :9.053e+09	Mean :198.4	Mean :2.917e+09	Mean : 67.23
3rd Qu.:1.528e+10	3rd Qu.:318.2	3rd Qu.:5.250e+09	3rd Qu.:120.10
Max. :2.310e+10	Max. :513.3	Max. :8.450e+09	Max. :197.10
NA's :14062	NA's :14062	NA's :14062	NA's :14062
Total_Debt	Annual_Income	Debt_to_Income_Ratio	
Min. :1.550e+09	Min. : 51216	Min. : 26696	
1st Qu.:2.075e+10	1st Qu.: 58172	1st Qu.: 350859	
Median :3.858e+10	Median : 61641	Median : 598880	
Mean :5.074e+10	Mean : 62512	Mean : 791169	
3rd Qu.:8.354e+10	3rd Qu.: 66450	3rd Qu.:1257181	

Max.	:1.340e+11	Max.	:100673	Max.	:1888695
NA's	:14062	NA's	:14062	NA's	:14062

2) Outlier Screening and Handling Procedures:

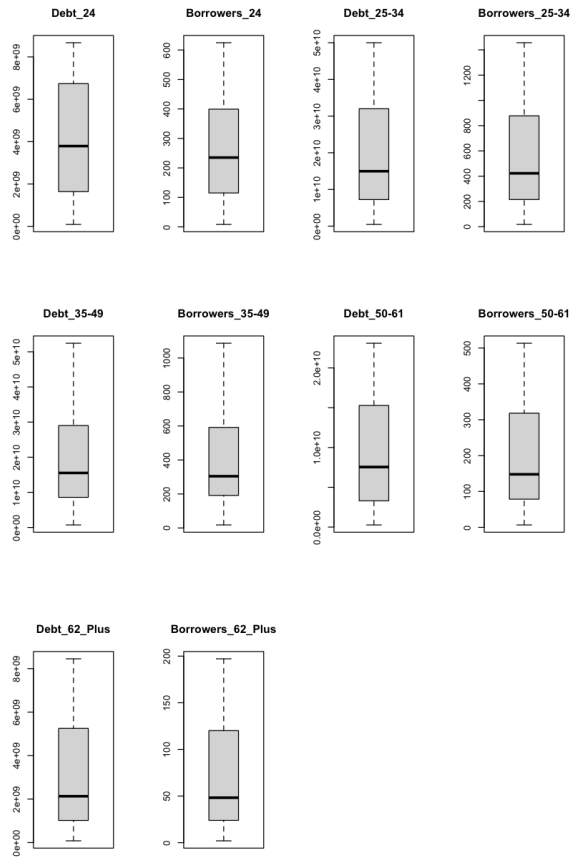
Inspect Debt Related Distributions:

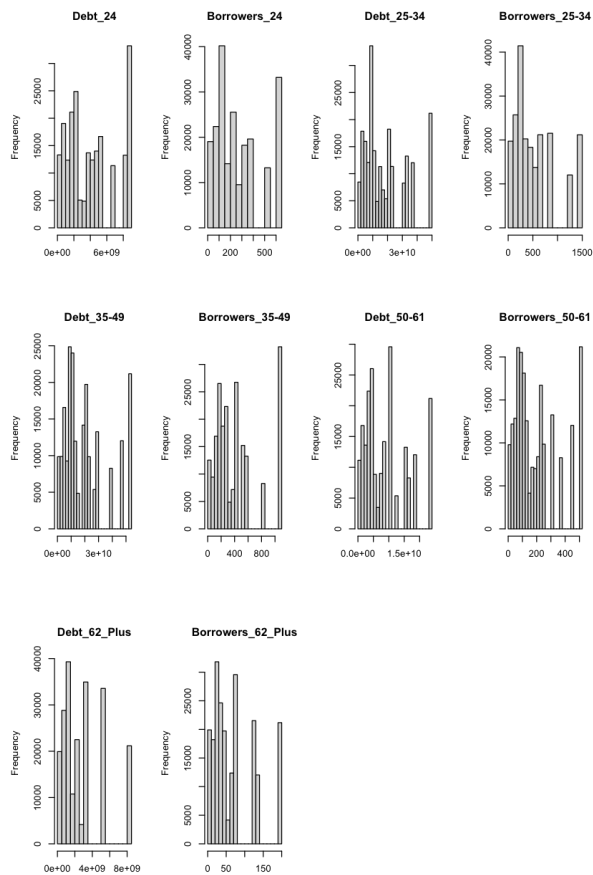
In [9]:

```
inspect_debt <- eda_df %>%
  select('Debt_24', 'Borrowers_24', 'Debt_25-34',
         'Borrowers_25-34', 'Debt_35-49', 'Borrowers_35-49', 'Debt_50-61', 'Borrowers_50-61', 'Debt_62_Plus', 'Borrowers_62_Plus')

# boxplots
par(mfrow = c(2,4))
for(col in names(inspect_debt))
  boxplot(inspect_debt[[col]], main = col)

# histogram
par(mfrow = c(2,4))
for(col in names(inspect_debt))
  hist(inspect_debt[[col]], main = col, xlab = "")
```





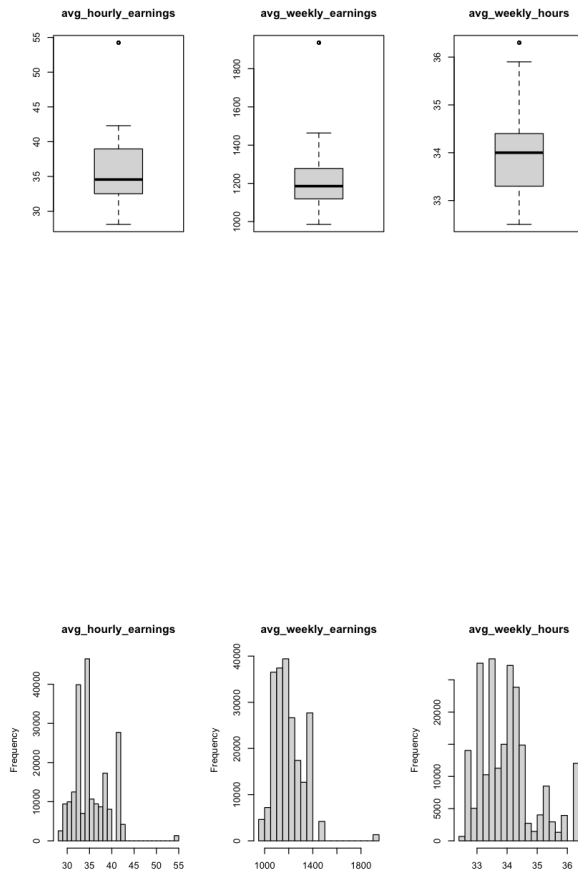
Interpretation:

The boxplots for debt and borrower-related variables show a rightward skew. This suggests outliers on the high end of the distribution. There is also variability in debt amounts across borrower age groups.

Inspect Income Related Distributions:

```
In [10]: inspect_income <- eda_df %>%
  select('avg_hourly_earnings', 'avg_weekly_earnings', 'avg_weekly_hours',)
# boxplots
par(mfrow = c(2,3))
for(col in names(inspect_income ))
  boxplot(inspect_income [[col]], main = col)

# histogram
par(mfrow = c(2,3))
for(col in names(inspect_income))
  hist(inspect_income[[col]], main = col, xlab = "")
```



Interpretation:

The distributions for income-related variables are relatively symmetric. There are some outliers, but they will be included in the analysis because they may describe a relationship between income and state.

Keep State Level Variables:

```
In [11]: analysis_df <- df

## All Columns:
# 'UNITID' 'OPEID6.x' 'INSTNM.x' 'CIPCODE' 'CIPDESC' 'CREDLEV' 'CREDDESC' 'OPEID' 'OPEID6.y' 'INSTNM.y' 'CITY' 'STABBR'
# 'CONTROL' 'REGION' 'TUITFTE' 'INEXPTE' 'TUITIONFEE_IN' 'TUITIONFEE_OUT' 'ROOMBOARD_OFF' 'OTHEREXPENSE_OFF' 'BOOKSUPPLY'
# 'UGDS' 'UGDS_MEN' 'UGDS_WOMEN' 'UGDS_WHITE' 'UGDS_BLACK' 'UGDS_HISP' 'UGDS_ASIAN' 'UGDS_AIAN' 'UGDS_NHPI' 'UGDS_2M'
# 'UGDS_NRA' 'UGDS_UNKN' 'avg_hourly_earnings' 'avg_weekly_earnings' 'avg_weekly_hours' 'Debt_24' 'Borrowers_24' 'Debt_25-34'
# 'Borrowers_25-34' 'Debt_35-49' 'Borrowers_35-49' 'Debt_50-61' 'Borrowers_50-61' 'Debt_62_Plus' 'Borrowers_62_Plus'
# 'Total_Debt' 'Annual_Income' 'Debt_to_Income_Ratio'

analysis_df <- analysis_df %>%
select(
  STABBR, # state
  CREDDESC,
  CONTROL,
  TUITFTE,
  TUITIONFEE_IN,
  TUITIONFEE_OUT,
  ROOMBOARD_OFF,
  OTHEREXPENSE_OFF,
  BOOKSUPPLY,
  Total_Debt,
  Annual_Income,
  Debt_to_Income_Ratio, # response
) %>%
filter(!is.na(STABBR), #filter NA
       !is.na(Debt_to_Income_Ratio))
```

Interpretation:

Aggregating by state will help us understand which factors explain differences in debt-to-income ratios by state.

It is essential to keep columns that connect at a state level. Taking the average of each predictor across states will help explain its influence on debt-to-income ratios.

Group by STABBR:

```
In [12]: analysis_df <- analysis_df %>%
group_by(STABBR) %>%
summarize(
  avg_ratio = mean(Debt_to_Income_Ratio, na.rm=TRUE),
  avg_tuition = mean(TUITFTE, na.rm=TRUE),
  avg_instate = mean(TUITIONFEE_IN, na.rm=TRUE),
  avg_outstate = mean(TUITIONFEE_OUT, na.rm=TRUE),
  avg_roomboard = mean(ROOMBOARD_OFF, na.rm=TRUE),
  avg_otherexp = mean(OTHEREXPENSE_OFF, na.rm=TRUE),
  avg_books = mean(BOOKSUPPLY, na.rm=TRUE),
  avg_totaldebt = mean(Total_Debt, na.rm=TRUE),
  avg_income = mean(Annual_Income, na.rm=TRUE)
)

head(analysis_df, 10)
```

A tibble: 10 × 10

STABBR	avg_ratio	avg_tuition	avg_instate	avg_outstate	avg_roomboard	avg_otherexp	avg_books	avg_totaldebt	avg_income
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AK	31485.13	8175.045	9232.635	21153.07	11322.273	5751.494	1429.635	2.1800e+09	69239.04
AL	368425.98	9485.071	11902.341	19264.00	9632.552	4485.413	1380.721	2.1410e+10	58112.08
AR	221452.91	5735.084	9098.558	13440.64	8684.535	5318.466	1277.136	1.1870e+10	53600.56
AZ	467354.86	7784.221	7535.006	16057.56	9278.634	5740.157	1289.798	2.9100e+10	62265.32
CA	1888694.74	9181.051	12270.178	20435.73	15831.749	4572.303	1374.194	1.3398e+11	70937.88
CO	388910.86	12733.212	15190.139	25936.07	11406.120	4016.558	1447.119	2.6510e+10	68164.72
CT	222798.99	12203.120	22891.384	34429.36	11727.411	3640.105	1080.936	1.5170e+10	68088.28
DC	60989.83	24820.725	41775.255	42273.87	14788.477	2559.454	1279.468	6.1400e+09	100672.52
DE	76483.97	13668.025	13763.057	24760.42	11337.102	2666.706	1416.331	4.2500e+09	55567.20
FL	1510188.30	9708.135	15063.253	21486.69	11023.471	4845.855	1267.453	9.3090e+10	61641.32

Group by STABBR and CREDESC:

```
In [13]: analysis_df_2 <- df %>%
group_by(STABBR, CREDESC) %>%
summarize(
  avg_ratio = mean(Debt_to_Income_Ratio, na.rm = TRUE),
  avg_tuition = mean(TUITFTE, na.rm = TRUE),
  avg_instate = mean(TUITIONFEE_IN, na.rm = TRUE),
  avg_outstate = mean(TUITIONFEE_OUT, na.rm = TRUE),
  avg_roomboard = mean(ROOMBOARD_OFF, na.rm = TRUE),
  avg_otherexp = mean(OTHEREXPENSE_OFF, na.rm = TRUE),
  avg_books = mean(BOOKSUPPLY, na.rm = TRUE),
  avg_totaldebt = mean(Total_Debt, na.rm = TRUE),
  avg_income = mean(Annual_Income, na.rm = TRUE),
  .groups = "drop"
)
analysis_df_2
```

A tibble: 449 × 11

STABBR	CREDESC	avg_ratio	avg_tuition	avg_instate	avg_outstate	avg_roomboard	avg_otherexp	avg_books	avg_totaldebt	avg_inc
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AK	Associate's Degree	31485.13	7691.829	9188.119	19723.136	10957.305	5870.271	1237.288	2.180e+09	6923
AK	Bachelor's Degree	31485.13	8499.537	9822.030	22215.591	11375.067	5907.250	1449.512	2.180e+09	6923
AK	Doctoral Degree	31485.13	8681.864	7924.909	21642.909	12099.364	4642.364	1901.818	2.180e+09	6923
AK	First Professional Degree	31485.13	9909.500	14691.000	22994.000	13684.000	6448.000	1260.000	2.180e+09	6923
AK	Graduate/Professional Certificate	31485.13	8613.054	9207.405	22816.541	11627.243	5926.865	1572.973	2.180e+09	6923
AK	Master's Degree	31485.13	8643.213	8892.303	22264.888	11764.888	5390.438	1654.831	2.180e+09	6923
AK	Post-baccalaureate Certificate	31485.13	9473.333	12186.000	22433.333	13139.333	5715.333	1506.667	2.180e+09	6923
	Undergraduate									

Interpretation:

Aggregating by state and degree type to explore potential differences in model performance due to the grouping of data.

Train and Test Data:

```
In [14]: set.seed(111)

# 70/30 split
train_split <- sample(1:nrow(analysis_df), size = floor(0.7 * nrow(analysis_df)))

trainData_prelim <- analysis_df[train_split, ]
testData_prelim <- analysis_df[-train_split, ]
```

Interpretation:

Creating two sets of training data. One grouped by State and one grouped by state and degree type.

Save Train and Test Data:

```
In [15]: trainData_prelim <- write_csv(trainData_prelim, "~/Documents/ISYE_6414_Final_Project-Adrian/1_data_loading_
```

```
In [16]: testData_prelim <- write_csv(testData_prelim, "~/Documents/ISYE_6414_Final_Project-Adrian/1_data_loading_csv_
```

```
In [17]: analysis_df_2 <- df %>%
  group_by(STABBR, CREDESC) %>%
  summarize(
    avg_ratio      = mean(Debt_to_Income_Ratio, na.rm = TRUE),
    avg_tuition    = mean(TUITFTE, na.rm = TRUE),
    avg_instate    = mean(TUITIONFEE_IN, na.rm = TRUE),
    avg_outstate   = mean(TUITIONFEE_OUT, na.rm = TRUE),
    avg_roomboard  = mean(ROOMBOARD_OFF, na.rm = TRUE),
    avg_otherexp   = mean(OTHEREXPENSE_OFF, na.rm = TRUE),
    avg_books      = mean(BOOKSUPPLY, na.rm = TRUE),
    avg_totaldebt  = mean(Total_Debt, na.rm = TRUE),
    avg_income     = mean(Annual_Income, na.rm = TRUE),
    .groups = "drop"
  )
analysis_df_2
```

A tibble: 449 × 11

STABBR	CREDESC	avg_ratio	avg_tuition	avg_instate	avg_outstate	avg_roomboard	avg_otherexp	avg_books	avg_totaldebt	avg_inc
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AK	Associate's Degree	31485.13	7691.829	9188.119	19723.136	10957.305	5870.271	1237.288	2.180e+09	6923
AK	Bachelor's Degree	31485.13	8499.537	9822.030	22215.591	11375.067	5907.250	1449.512	2.180e+09	6923
AK	Doctoral Degree	31485.13	8681.864	7924.909	21642.909	12099.364	4642.364	1901.818	2.180e+09	6923
AK	First Professional Degree	31485.13	9909.500	14691.000	22994.000	13684.000	6448.000	1260.000	2.180e+09	6923
AK	Graduate/Professional Certificate	31485.13	8613.054	9207.405	22816.541	11627.243	5926.865	1572.973	2.180e+09	6923
AK	Master's Degree	31485.13	8643.213	8892.303	22264.888	11764.888	5390.438	1654.831	2.180e+09	6923
AK	Post-baccalaureate Certificate	31485.13	9473.333	12186.000	22433.333	13139.333	5715.333	1506.667	2.180e+09	6923
Undergraduate										

```
In [18]: set.seed(111)

# 70/30 split
train_split <- sample(1:nrow(analysis_df_2), size = floor(0.7 * nrow(analysis_df_2)))

trainData <- analysis_df_2 [train_split, ]
testData <- analysis_df_2 [-train_split, ]
```

```
In [19]: trainData <- write_csv(trainData, "~/Documents/ISYE_6414_Final_Project-Adrian/1_data_loading_cleaning/merged_trainData.csv")
```

```
In [20]: testData <- write_csv(testData, "~/Documents/ISYE_6414_Final_Project-Adrian/1_data_loading_cleaning/merged_testData.csv")
```

3) Variable Selection Procedures:

Preliminary Model: model1

Predictors:

avg_tuition
avg_instate
avg_outstate
avg_roomboard
avg_otherexp
avg_books
avg_totaldebt
avg_income

Grouped by:

State (STABBR)

Note:

This model includes all predictors

In [21]:

```
model1 <- lm(avg_ratio ~ avg_tuition + avg_instate + avg_outstate + avg_roomboard +
             avg_otherexp + avg_books + avg_totaldebt + avg_income, data = trainData_prelim)

summary(model1)
```

Call:

```
lm(formula = avg_ratio ~ avg_tuition + avg_instate + avg_outstate +
    avg_roomboard + avg_otherexp + avg_books + avg_totaldebt +
    avg_income, data = trainData_prelim)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-75718 -18135  -323    10391   69764
```

Coefficients:

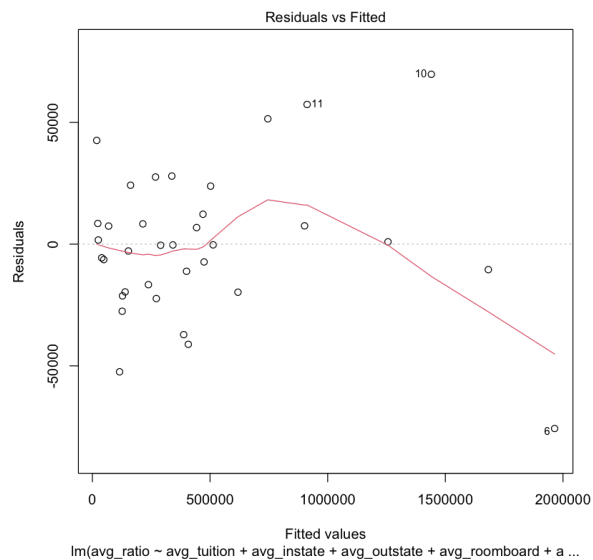
```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.673e+05  7.924e+04   2.112  0.04446 *
avg_tuition    9.478e-01  2.771e+00   0.342  0.73505
avg_instate    2.967e+00  2.384e+00   1.244  0.22445
avg_outstate   -6.158e-01  2.727e+00  -0.226  0.82313
avg_roomboard  -1.629e+01  5.779e+00  -2.819  0.00908 **
avg_otherexp    9.997e+00  8.340e+00   1.199  0.24149
avg_books      5.407e+01  4.896e+01   1.104  0.27953
avg_totaldebt  1.535e-05  2.264e-07  67.780 < 2e-16 ***
avg_income    -2.170e+00  1.040e+00  -2.086  0.04691 *
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 34230 on 26 degrees of freedom
Multiple R-squared:  0.9961,    Adjusted R-squared:  0.9949
F-statistic: 834.1 on 8 and 26 DF,  p-value: < 2.2e-16
```

In [22]:

```
plot(model1, which = 1)
```

**Interpretation:**

Model 1: This served as a preliminary model, using all predictors and the grouped-by-state dataset. This model produced a suspiciously high R-squared of 0.9961, suggesting multicollinearity. The adjusted R-squared penalizes the number of predictors, yet it is also suspiciously high. The predictors appear to be correlated with one another. A potential problem is that the response variable (debt-to-income ratio) is calculated using predictors that capture average income and average debt.

Reduced Model: model2**Predictors:**

avg_totaldebt

avg_income

Grouped by:

State (STABBR)

```
In [23]: model2 <- lm(avg_ratio ~ avg_totaldebt + avg_income, data = trainData_prelim)
summary(model2)
```

Call:

```
lm(formula = avg_ratio ~ avg_totaldebt + avg_income, data = trainData_prelim)
```

Residuals:

Min	1Q	Median	3Q	Max
-132117	-15529	-7521	18428	78339

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.192e+05	4.918e+04	4.457	9.55e-05 ***
avg_totaldebt	1.512e-05	2.138e-07	70.724	< 2e-16 ***
avg_income	-3.167e+00	7.774e-01	-4.074	0.000285 ***

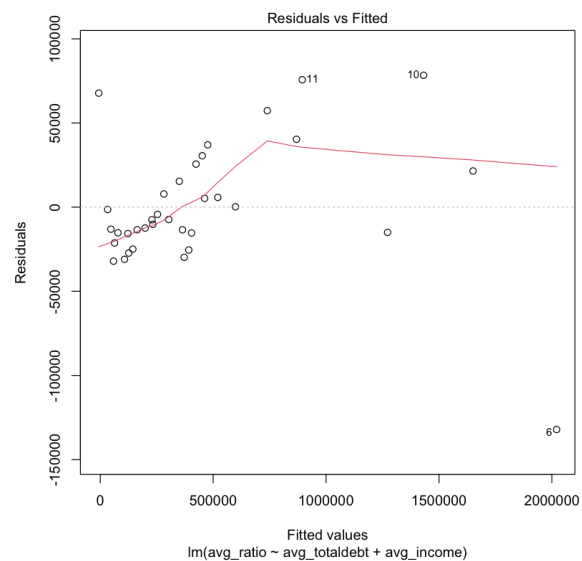
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39450 on 32 degrees of freedom

Multiple R-squared: 0.9937, Adjusted R-squared: 0.9933

F-statistic: 2505 on 2 and 32 DF, p-value: < 2.2e-16

```
In [24]: plot(model2, which = 1)
```



Interpretation:

Model 2: To confirm suspicions about the relationship between response and predictors in Model 1, this model only included the predictors used to calculate the response variable. As expected, there was a strong relationship. This model also exemplified a high R-squared value. This model does not uncover any new relationships between the predictors and response.

Reduced Preliminary Model: model3

Predictors:

avg_tuition

avg_instate

avg_outstate

avg_roomboard

avg_otherexp

avg_books

Grouped by:

State (STABBR)

Note: Removed predictors contributing to the calculation of the debt-to-income ratio (response variable).

```
In [25]: model3 <- lm(avg_ratio ~ avg_tuition + avg_instate + avg_outstate + avg_roomboard +  
  avg_otherexp + avg_books, data = trainData_prelim)  
summary(model3)
```

Call:

```
lm(formula = avg_ratio ~ avg_tuition + avg_instate + avg_outstate +  
  avg_roomboard + avg_otherexp + avg_books, data = trainData_prelim)
```

Residuals:

Min	1Q	Median	3Q	Max
-737997	-229441	-80895	114562	1228864

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-718705.37	786052.55	-0.914	0.3684
avg_tuition	-36.88	35.09	-1.051	0.3021
avg_instate	41.59	28.61	1.454	0.1572
avg_outstate	-36.47	35.37	-1.031	0.3114
avg_roomboard	134.72	54.07	2.492	0.0189 *
avg_otherexp	89.34	109.43	0.816	0.4211
avg_books	12.87	601.67	0.021	0.9831

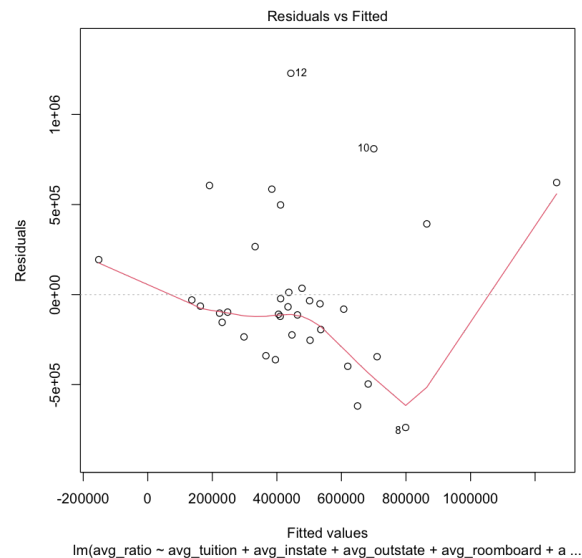
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 454400 on 28 degrees of freedom

Multiple R-squared: 0.2631, Adjusted R-squared: 0.1052

F-statistic: 1.666 on 6 and 28 DF, p-value: 0.1663

```
In [26]: plot(model3, which = 1)
```



Interpretation:

Model 3: To fully confirm suspicions of multicollinearity raised in Models 1 and 2, this approach tests all predictors except those used to create the response. One thing predictors have in common is that they are related to a cost, for instance, tuition, books, room and board, etc. The r-squared value was 0.2631, meaning that this combination of predicting variables explains about 26% of the variation in the response variable. The adjusted R-squared value, however, was 0.1052. This suggested that cost predictors alone did not display strong predictive power.

Grouped by State and Institution Type: model4

Predictors:

avg_tuition
avg_instate
avg_outstate
avg_roomboard
avg_otherexp
avg_books
avg_totaldebt
avg_income
CREDESC

Grouped by:

State(STABBR) CREDESC (School Type)

Note: Removed predictors contributing to the calculation of the debt to income ratio (response variable)

```
In [27]: model4 <- lm(
  avg_ratio ~ avg_tuition + avg_instate + avg_outstate + avg_roomboard +
    avg_otherexp + avg_books + avg_totaldebt + avg_income + CREDDDESC , data = trainData)
summary(model4)
```

Call:

```
lm(formula = avg_ratio ~ avg_tuition + avg_instate + avg_outstate +
  avg_roomboard + avg_otherexp + avg_books + avg_totaldebt +
  avg_income + CREDDDESC, data = trainData)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-148048	-15500	-1976	14626	85109

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	2.403e+05	2.596e+04	9.258
avg_tuition	-1.730e-01	2.440e-01	-0.709
avg_instate	1.703e+00	5.235e-01	3.252
avg_outstate	3.133e-01	6.225e-01	0.503
avg_roomboard	-8.680e+00	1.615e+00	-5.375
avg_otherexp	3.728e+00	2.544e+00	1.466
avg_books	2.630e+01	1.259e+01	2.089
avg_totaldebt	1.532e-05	8.178e-08	187.393
avg_income	-3.372e+00	3.520e-01	-9.580
CREDDDESCBachelor's Degree	-8.614e+03	1.004e+04	-0.858
CREDDDESCDoctoral Degree	2.755e+03	1.166e+04	0.236
CREDDDESCFirst Professional Degree	-1.968e+03	1.081e+04	-0.182
CREDDDESCGraduate/Professional Certificate	-2.458e+03	1.022e+04	-0.240
CREDDDESCMaster's Degree	-3.746e+03	1.015e+04	-0.369
CREDDDESCPost-baccalaureate Certificate	-2.561e+03	1.016e+04	-0.252
CREDDDESCUndergraduate Certificate or Diploma	5.692e+03	8.230e+03	0.692

	Pr(> t)
(Intercept)	< 2e-16 ***
avg_tuition	0.47909
avg_instate	0.00129 **
avg_outstate	0.61513
avg_roomboard	1.68e-07 ***
avg_otherexp	0.14391
avg_books	0.03769 *
avg_totaldebt	< 2e-16 ***
avg_income	< 2e-16 ***
CREDDDESCBachelor's Degree	0.39183
CREDDDESCDoctoral Degree	0.81345
CREDDDESCFirst Professional Degree	0.85569
CREDDDESCGraduate/Professional Certificate	0.81016
CREDDDESCMaster's Degree	0.71245
CREDDDESCPost-baccalaureate Certificate	0.80121
CREDDDESCUndergraduate Certificate or Diploma	0.48982

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

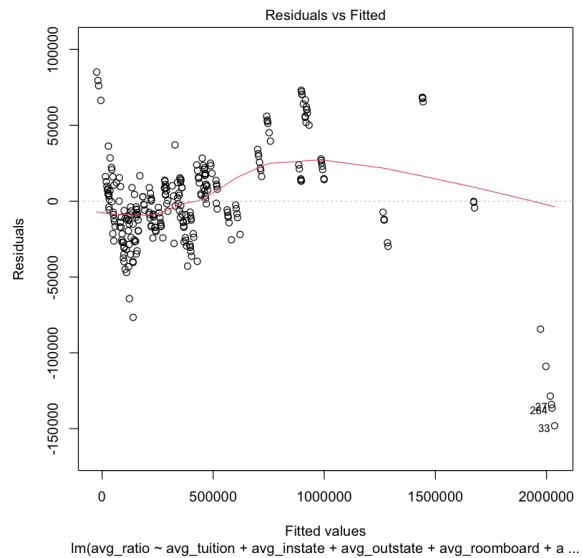
Residual standard error: 33850 on 265 degrees of freedom

(33 observations deleted due to missingness)

Multiple R-squared: 0.9939, Adjusted R-squared: 0.9936

F-statistic: 2890 on 15 and 265 DF, p-value: < 2.2e-16

```
In [28]: plot(model4, which = 1)
```



Interpretation:

Model 4: This model emulates Model 1 by using all predictors except those used to calculate the response variable. This model uses data grouped by state and degree type. Similar to Model 2, it shows a very high R-squared value, reflecting the strong relationship between the predictors and the response.

Reduced Model: model5

Predictors:

avg_tuition
avg_instate
avg_outstate
avg_roomboard
avg_otherexp
avg_books
CREDESC

Grouped by:

State(STABBR) CREDESC (School Type)

Note:

Removed predictors contributing to the calculation of the debt-to-income ratio (response variable).

```
In [29]: model5 <- lm(
  avg_ratio ~ avg_tuition + avg_instate + avg_outstate + avg_roomboard +
    avg_otherexp + avg_books + CREDDDESC, data = trainData)
summary(model5)
```

Call:

```
lm(formula = avg_ratio ~ avg_tuition + avg_instate + avg_outstate +
  avg_roomboard + avg_otherexp + avg_books + CREDDDESC, data = trainData)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-873711 -215073  -72488   141153 1241500
```

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	-4.414e+05	2.263e+05	-1.950
avg_tuition	-5.691e+00	2.783e+00	-2.045
avg_instate	2.284e+01	5.778e+00	3.953
avg_outstate	-8.390e+00	7.181e+00	-1.168
avg_roomboard	5.191e+01	1.647e+01	3.152
avg_otherexp	9.099e+01	2.893e+01	3.145
avg_books	4.045e+00	1.446e+02	0.028
CREDDDESCBachelor's Degree	-2.831e+05	1.119e+05	-2.530
CREDDDESCDoctoral Degree	-8.254e+04	1.307e+05	-0.631
CREDDDESCFirst Professional Degree	-2.090e+05	1.212e+05	-1.725
CREDDDESCGraduate/Professional Certificate	-1.682e+05	1.146e+05	-1.468
CREDDDESCMaster's Degree	-1.284e+05	1.139e+05	-1.128
CREDDDESCPost-baccalaureate Certificate	-1.661e+05	1.147e+05	-1.447
CREDDDESCUndergraduate Certificate or Diploma	-8.122e+04	9.506e+04	-0.854

	Pr(> t)
(Intercept)	0.05218 .
avg_tuition	0.04186 *
avg_instate	9.89e-05 ***
avg_outstate	0.24368
avg_roomboard	0.00181 **
avg_otherexp	0.00185 **
avg_books	0.97771
CREDDDESCBachelor's Degree	0.01199 *
CREDDDESCDoctoral Degree	0.52830
CREDDDESCFirst Professional Degree	0.08577 .
CREDDDESCGraduate/Professional Certificate	0.14325
CREDDDESCMaster's Degree	0.26054
CREDDDESCPost-baccalaureate Certificate	0.14898
CREDDDESCUndergraduate Certificate or Diploma	0.39368

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

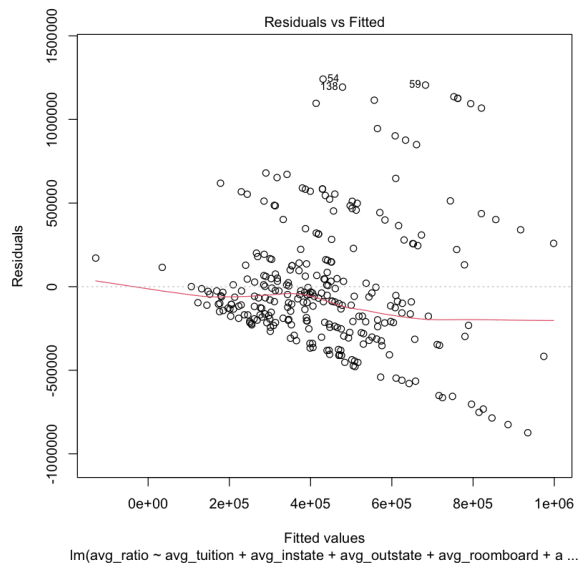
Residual standard error: 391600 on 267 degrees of freedom

(33 observations deleted due to missingness)

Multiple R-squared: 0.1807, Adjusted R-squared: 0.1408

F-statistic: 4.529 on 13 and 267 DF, p-value: 5.401e-07

```
In [30]: plot(model5, which = 1)
```



Interpretation:

Model 5: This model uses cost-related predictors, except those used to calculate the response variable. In addition, a new predictor, CREDDDES, is introduced, which describes the degree to which an institution grants. This model has an R-squared value of 0.1807, indicating that it explains about 18% of the variation in the data. In addition, the adjusted R-squared does not drop significantly when penalizing for the number of predictors.

This model is used as the base model for this analysis, given its reasonable R-squared value and its statistically significant predictors. Model 5 showed that cost-related variables were not statistically significant. This reduced the predictors to 4 main predicting variables: average tuition, average in-state tuition, average room and board, and average other expenses.

Summary:

Model 1: A preliminary model that includes all predictors, including avg_totaldebt and avg_income, which makes it invalid for this analysis.

Model 2: Uses avg_totaldebt and avg_income as predictors. However, avg_totaldebt and avg_income are used to create the response variable. This is an invalid approach given that the relationships in the predictors directly influence the response variable. This results in multicollinearity.

Model 3: Demonstrates an R-squared value of 0.2631, indicating that this set of predictor variables explains about 26% of the variation in the response variable. However, the adjusted R-squared value was 0.1052. This suggested that cost predictors alone did not display strong predictive power.

Model 4: A valid approach, but it may be multicollinear.

Model 5: Given the goal of identifying which factors influence the debt-to-income ratio at the state level, Model 5 is the most appropriate. Model 5 does not include avg_totaldebt and avg_income; it includes tuition-related and institution variables. Model 5 offers a great starting point. This model is built on variables relating to costs, such as avg_tuition, avg_instate, avg_outstate, avg_roomboard, and avg_books, as well as a dedicated variable that tracks institution type (CREDDDESC).

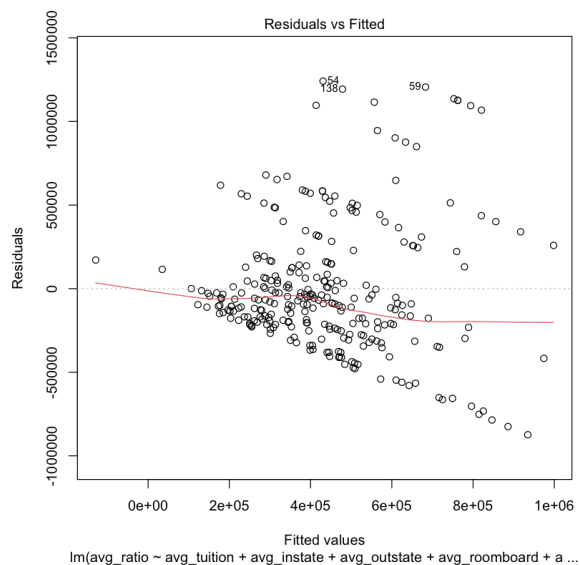
4) Goodness of Fit Testing and Model Assumption Checks:

Regression Assumptions:

1. Linearity
2. Constant Variance
3. Normality
4. Independent Errors

1) Check for Linearity:


```
In [31]: plot(model5, which = 1)
```

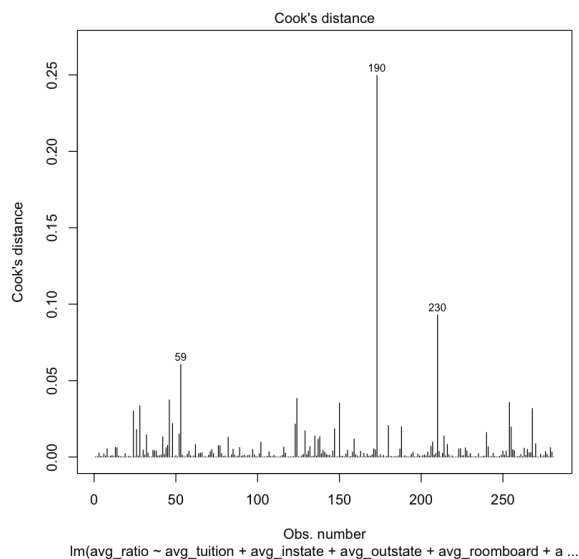


Interpretation:

Linearity is violated. The residuals demonstrate a slight curve. The spread of the residuals in larger values suggests heteroscedasticity. Given that the residuals are not centered around zero, the relationship between the predictors and the response variable is nonlinear. To improve linearity, I will identify and remove influential points below.

Cook's Distance:

```
In [32]: plot(model5, which = 4)
# Helps to see if outliers are skewing results.
```



```
In [33]: cooks_distance <- cooks.distance(model5)
which(cooks_distance > 4/length(cooks_distance)) # Module 2
```

27: 24 31: 26 33: 28 38: 32 52: 46 54: 48 58: 52 59: 53 138: 123 139: 124 144: 129 163: 147 166: 150 190: 173 198: 180 206: 188 230: 210 266: 240 284: 254 285: 255 298: 268

Remove Influential Points:

```
In [34]: cooks_distance <- cooks.distance(model5)
influ_points <- which(cooks_distance > 0.15) # 4/n is too strict, 0.15 will remove strong influences
influ_points
```

190: 173

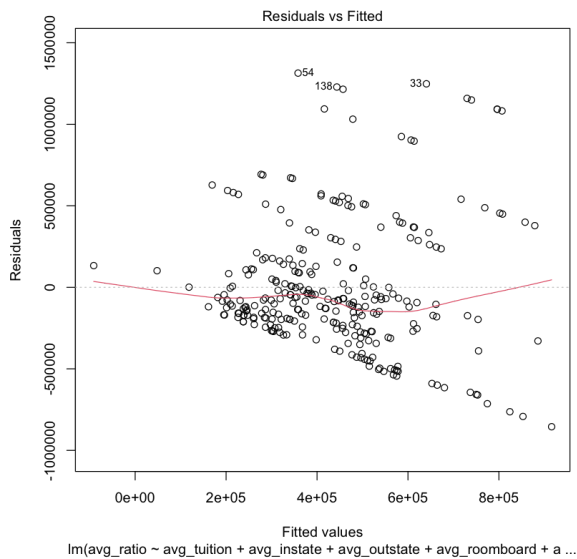
Interpretation

To improve linearity, I have used Cook's distance to identify potential influential points and used the threshold of 0.15 to remove strong influences.

```
In [35]: train_clean <- trainData[-influ_points, ] # data without influential points

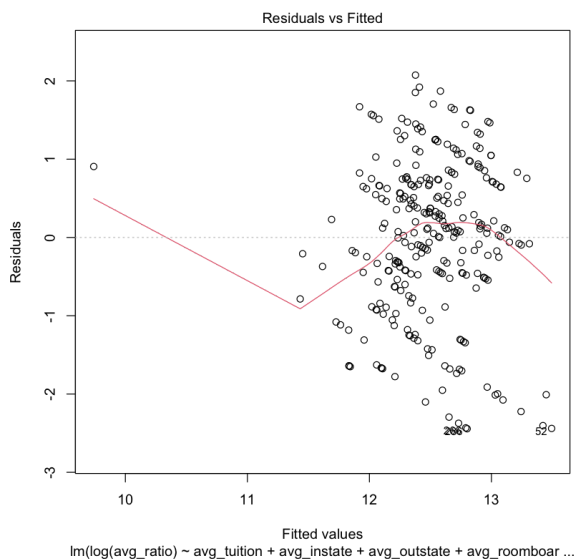
model5_clean <- lm(avg_ratio ~ avg_tuition + avg_instate + avg_outstate +
                  avg_roomboard + avg_otherexp + avg_books, data = train_clean)

plot(model5_clean, which = 1)
```



Apply Log Transformation:

```
In [36]: model_log <- lm(log(avg_ratio) ~ avg_tuition + avg_instate + avg_outstate +
                        avg_roomboard + avg_otherexp + avg_books,
                        data=train_clean)
plot(model_log, which=1)
```



```
In [37]: summary(model_log)
```

```
Call:
lm(formula = log(avg_ratio) ~ avg_tuition + avg_instate + avg_outstate +
    avg_roomboard + avg_otherexp + avg_books, data = train_clean)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.4434 -0.5233  0.1022  0.7176  2.0739
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.071e+01  5.577e-01  19.201  < 2e-16 ***
avg_tuition   -2.233e-05  7.070e-06  -3.158  0.00177 **
avg_instate    5.848e-05  1.379e-05   4.242  3.04e-05 ***
avg_outstate  -1.221e-05  1.452e-05  -0.841  0.40104
avg_roomboard  4.876e-06  4.198e-05   0.116  0.90762
avg_otherexp   2.693e-04  7.092e-05   3.797  0.00018 ***
avg_books      2.293e-04  3.497e-04   0.656  0.51264
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.008 on 273 degrees of freedom
(33 observations deleted due to missingness)
Multiple R-squared:  0.1393,    Adjusted R-squared:  0.1204
F-statistic: 7.366 on 6 and 273 DF,  p-value: 2.576e-07
```

Interpretation:

Keep only variables that are statistically significant. avg_tuition, avg_tuition, and avg_otherexp are statistically significant.

```
In [38]: model_log_sig <- lm(log(avg_ratio) ~ avg_tuition + avg_instate + avg_otherexp,
                             data = train_clean)
summary(model_log_sig)
```

```
Call:
lm(formula = log(avg_ratio) ~ avg_tuition + avg_instate + avg_otherexp,
    data = train_clean)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.45123 -0.57902  0.08934  0.72298  2.19322
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.088e+01  3.446e-01  31.570  < 2e-16 ***
avg_tuition   -2.402e-05  6.761e-06  -3.553  0.000448 ***
avg_instate    4.868e-05  8.249e-06   5.901  1.05e-08 ***
avg_otherexp   2.804e-04  6.853e-05   4.092  5.63e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.005 on 276 degrees of freedom
(33 observations deleted due to missingness)
Multiple R-squared:  0.1343,    Adjusted R-squared:  0.1249
F-statistic: 14.27 on 3 and 276 DF,  p-value: 1.139e-08
```

Interpretation:

I simplified model_log and created model_log_sig which is composed of predictors that demonstrated statistical significance.

To address the violation of the linearity assumption, I applied a log transformation to flatten the scale to make the linearity assumption more apparent (similar to the log transformation applied on the Palm Beach County example in class). The linearity assumption is better, the errors are scattered around 0, and the megaphone shape has subsided. I will first check for constant variance and normality below.

2) Check for Constant Variance:

In [39]:

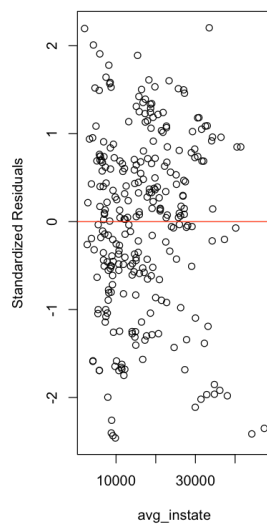
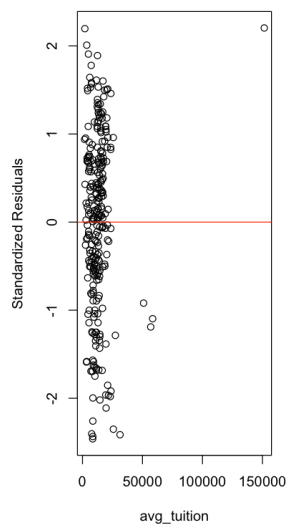
```
df_used <- model.frame(model_log_sig) # get model fram from model_log_sig

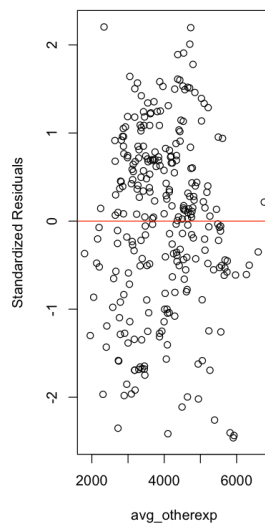
model_log_sig_resids <- rstandard(model_log_sig)

par(mfrow = c(1, 2))
plot(df_used$avg_tuition, model_log_sig_resids,
     xlab = "avg_tuition", ylab = "Standardized Residuals")
abline(h = 0, col = "red")

par(mfrow = c(1, 2))
plot(df_used$avg_instate, model_log_sig_resids,
     xlab = "avg_instate", ylab = "Standardized Residuals")
abline(h = 0, col = "red")

par(mfrow = c(1, 2))
plot(df_used$avg_otherexp, model_log_sig_resids,
     xlab = "avg_otherexp", ylab = "Standardized Residuals")
abline(h = 0, col = "red")
```





Interpretation:

Constant variance assumption is met. Avg_tuition demonstrated a consistent spread from -2 to 2 with no megaphone or funnel-shaped patterns. Avg_instate demonstrated a consistent spread without any patterns. Avg_otherexp showed a consistent spread with no patterns. All three predictors pass the Constant Variance Assumption; therefore, errors demonstrate homoscedasticity.

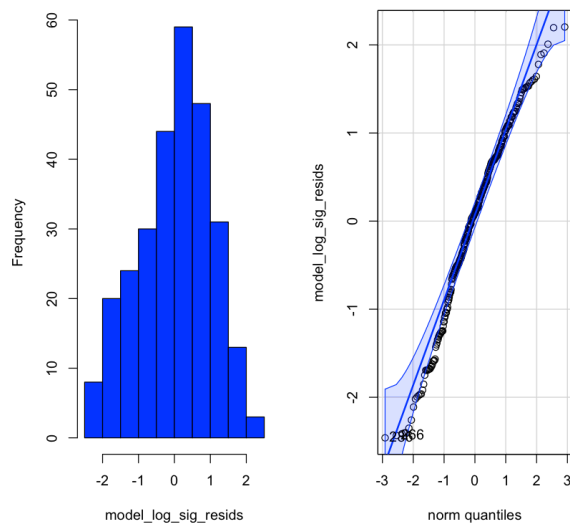
3) Normality:

Scatterplot of the Standardized Residuals vs. the Fitted Values of model1:

```
In [40]: # QQ plot.
par(mfrow = c(1, 2))
hist(model_log_sig_resids, col = "blue")
qqPlot(model_log_sig_resids, envelope = 0.95)
```

286: 256 266: 240

Histogram of model_log_sig_resid



Interpretation:

The residuals from the log-transformed model exhibit a normal distribution, as shown by the bell-shaped distribution in the barplot. The QQ plot shows that most points are within the shaded region.

Multicollinearity:

```
In [41]: # VIF threshold.  
cat("VIF threshold: ", max(10, 1/(1-summary(model_log_sig)$r.squared))), "\n")  
# VIF.  
vif(model_log_sig)
```

VIF threshold: 10

avg_tuition: 1.48532730402555 avg_instate: 1.61404919697947 avg_otherexp: 1.19687174552079

Interpretation:

To assess multicollinearity among predictors, I ran a VIF test. Multicollinearity occurs when variables are too closely related and therefore demonstrate the same relationship with the response variable. VIF is a way to determine whether multicollinearity is strong using a threshold of 2. Since all predictor variables are below 2, multicollinearity is not an issue.

ANOVA:

```
In [42]: anova(model_log_sig)
```

A anova: 4 × 5

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
avg_tuition	1	1.935645	1.935645	1.915119	1.675138e-01
avg_instate	1	24.406031	24.406031	24.147227	1.532211e-06
avg_otherexp	1	16.920714	16.920714	16.741285	5.627912e-05
Residuals	276	278.958097	1.010718	NA	NA

Interpretation:

According to the ANOVA test avg_tuition is not significant when all predictors are in the model. ANOVA suggests that avg_tuition is collinear with avg_instate.

```
In [43]: car::vif(model_log_sig)
```

avg_tuition: 1.48532730402555 avg_instate: 1.61404919697947 avg_otherexp: 1.19687174552079

Given that all VIF values are below the threshold of 2, there is no colinearity, but avg_instate is a weaker predictor.

F-test:

Given that the f value is relatively large at 14.27, it suggests that the predictors avg_tuition, avg_instate, and avg_other explain variation in the model.

P-value:

The p-value is low at 1.139e-08. This suggests that the model is statistically significant, implying that it is unlikely that the predictor does NOT influence the response variable.

R-squared:

The R-squared values are relatively low. At 0.1343, the R-squared value explains about 13 percent of the variance in the data. While the model is statistically significant based on its p-value, there are other factors unaccounted for in this model.

5) Model training and Evaluation Code:

```
In [44]: model_log_sig2 <- lm(log(avg_ratio) ~ avg_instate + avg_otherexp, data = train_clean) ### Simplify the model
summary(model_log_sig2)
```

Call:

```
lm(formula = log(avg_ratio) ~ avg_instate + avg_otherexp, data = train_clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.4890	-0.6133	0.1193	0.7635	2.2202

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.073e+01	3.493e-01	30.732	< 2e-16 ***
avg_instate	3.362e-05	7.223e-06	4.654	5.04e-06 ***
avg_otherexp	3.017e-04	6.968e-05	4.330	2.09e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.026 on 277 degrees of freedom

(33 observations deleted due to missingness)

Multiple R-squared: 0.09467, Adjusted R-squared: 0.08814

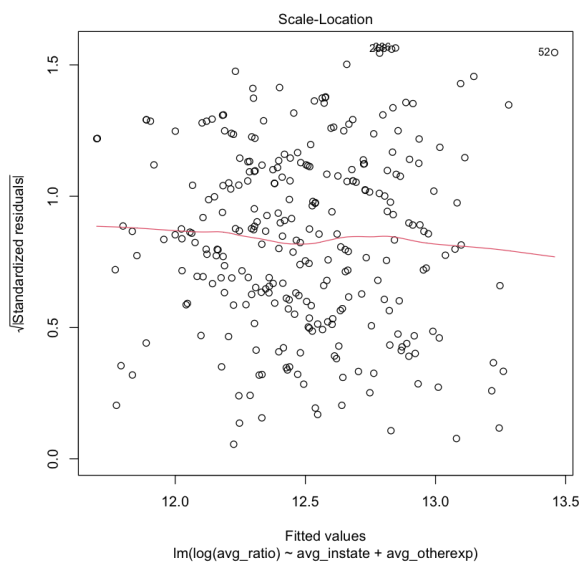
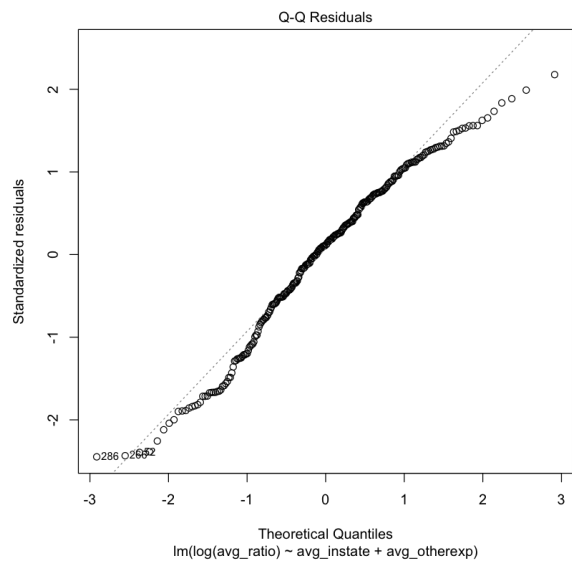
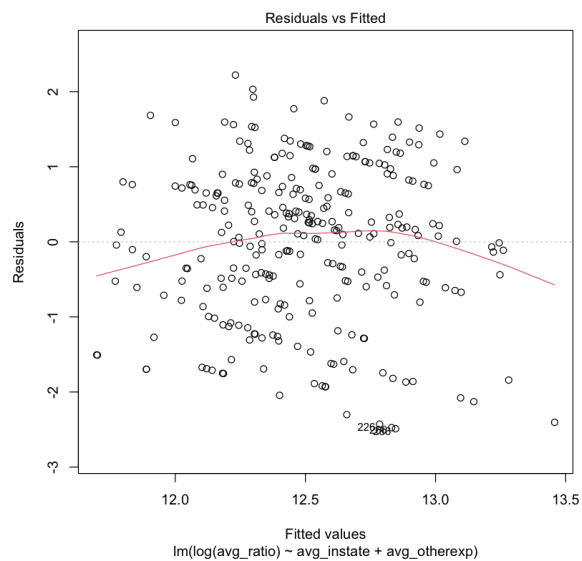
F-statistic: 14.48 on 2 and 277 DF, p-value: 1.041e-06

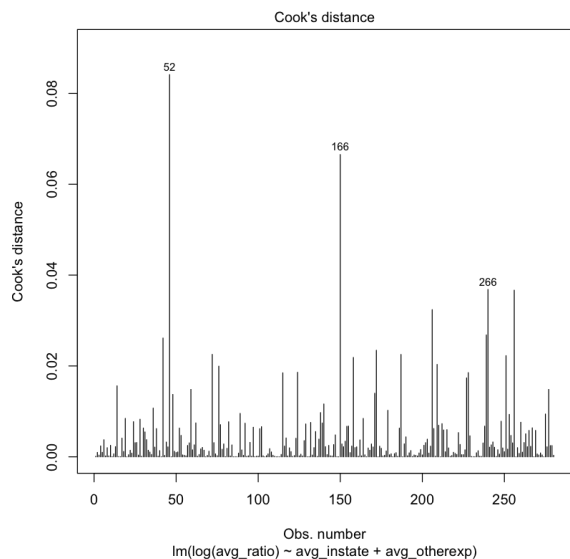
Interpretation:

The ANOVA test demonstrated that avg_tuition was not a significant predictor when paired with avg_instate and avg_otherexp. To simplify the model, I removed avg_tuition.

I will explore this as model_log_sig2. Below, I will investigate the core assumptions relating to this reduced model.

```
In [45]: plot(model_log_sig2, which=1:4)
```





Interpretation:

All regression assumptions are met under this reduced model; however, I will use cross-validation to help choose a final model.

6) Statistical Analysis and Hypothesis Testing:

I cannot apply deviance and pearson goodness of fit test to Multiple linear regression models.

Hypothesis Testing:

Null Hypothesis: The null hypothesis is that the predictors avg_tuition, avg_instate, and avg_otherexp do not have a relationship with the response variable log(avg_ratio).

Alternative Hypothesis: The alternative hypothesis is that the predictors avg_tuition, avg_instate, and avg_otherexp do indeed have a relationship with the response variable log(avg_ratio).

Given that the p value is less than 0.05, it is reasonable to conclude that the predictors help predict the response. There the null hypothesis is rejected.

7) Cross-validation and Model Selection:

Need to get rid of NA values in response variable

```
In [46]: colSums(is.na(train_clean))
sum(is.na(train_clean$avg_ratio))
```

```
STABBR: 6 CREDESC: 0 avg_ratio: 33 avg_tuition: 6 avg_instate: 6 avg_outstate: 6 avg_roomboard: 6 avg_otherexp: 6 avg_books: 6
avg_totaldebt: 33 avg_income: 33
```

```
33
```

Interpretation:

There are missing values in response and predictors. Simple solution is to drop NA values.

```
In [47]: trainData_cv <- na.omit(train_clean)
```

```

In [48]: library(caret)

train_control <- trainControl(method = "cv", number = 5)

# filtered with only significant values and ANOVA suggestion
cv_model_log_sig2 <- train( # only 2 predictors (takes into account ANOVA )
  formula(model_log_sig2),
  data = trainData_cv,
  method = "lm",
  trControl = train_control)

# filtered with only significant values
cv_model_log_sig <- train( # 3 predictor
  formula(model_log_sig),
  data = trainData_cv,
  method = "lm",
  trControl = train_control)

# unreduced model
cv_model_log <- train(
  formula(model_log),
  data = trainData_cv,
  method = "lm",
  trControl = train_control)

cv_model_log_sig2
cv_model_log_sig
cv_model_log

```

Linear Regression

280 samples
2 predictor

No pre-processing

Resampling: Cross-Validated (5 fold)

Summary of sample sizes: 224, 223, 225, 224, 224

Resampling results:

RMSE	Rsquared	MAE
1.024549	0.09391303	0.8366988

Tuning parameter 'intercept' was held constant at a value of TRUE

Linear Regression

280 samples
3 predictor

No pre-processing

Resampling: Cross-Validated (5 fold)

Summary of sample sizes: 224, 223, 225, 224, 224

Resampling results:

RMSE	Rsquared	MAE
1.021152	0.1164187	0.8207335

Tuning parameter 'intercept' was held constant at a value of TRUE

Linear Regression

280 samples
6 predictor

No pre-processing

Resampling: Cross-Validated (5 fold)

Summary of sample sizes: 224, 223, 225, 224, 224

Resampling results:

RMSE	Rsquared	MAE
1.071933	0.1078498	0.8378718

Tuning parameter 'intercept' was held constant at a value of TRUE

Interpretation:

To assess the model's performance on unseen data, I conducted 5-fold cross-validation. The base model is model_log, which applies a log transformation to its response variable (debt-to-income ratio) and averages it by state (avg_ratio).

model_log_sig is similar to model_log, but includes only predictors deemed statistically significant after reviewing p-values at the 0.05 significance threshold.

model_log_sig2 has only two predictors; it uses p-value-based tests and ANOVA results to justify the reduction in variables.

While model_log_sig2 has the lowest RMSE of 1.027231, model_log_sig still better explains the data's variance. model_log_sig explains about 13% of the variance in the data.

```
In [49]: final_model <- model_log_sig
```

model_log_sig is the final model.

```
In [50]: # remove rows with missing values
testData <- na.omit(testData)

# prediction
predicted_log <- predict(final_model, newdata = testData)

predicted_ratio <- exp(predicted_log) # convert predictions back to debt-to-income ratio, since response is
```

```
In [51]: head(predicted_ratio,5)
```

1: 357423.488447512 2: 520884.248068872 3: 379415.318849273 4: 281090.277741106 5: 273815.047413638

```
In [52]: summary(trainData$avg_ratio)
summary(testData$avg_ratio)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
26696	119448	341891	438278	557097	1888695	33
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
26696	98995	295987	415810	512882	1888695	

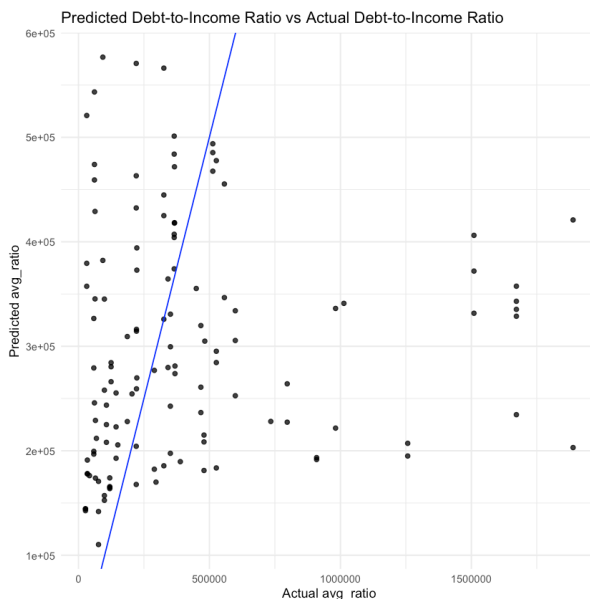
Interpretation:

- Values are with in same magnitude.

Extra Visualizations:

```
In [53]: testData$predicted_ratio <- predicted_ratio

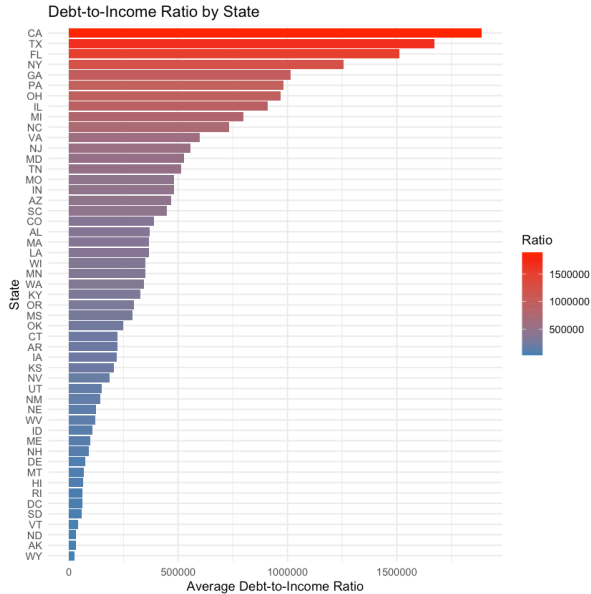
ggplot(testData, aes(x = avg_ratio, y = predicted_ratio)) + geom_point(alpha = 0.8) +
geom_abline(slope = 1, intercept = 0, color = "blue") +
labs(title = "Predicted Debt-to-Income Ratio vs Actual Debt-to-Income Ratio",
x = "Actual avg_ratio",
y = "Predicted avg_ratio") + theme_minimal()
```



Interpretation:

The model created in this analysis was moderate at predicting low- to mid-sized values but performed poorly at predicting larger values.

```
In [54]: trainData_cv %>% # cv dataframe removed NA columns
group_by(STABBR) %>% # State Abbreviation column will function as state column
summarize(mean_ratio = mean(avg_ratio)) %>%
ggplot(aes(x = reorder(STABBR, mean_ratio), y = mean_ratio, fill = mean_ratio)) +
geom_col() +
coord_flip() +
scale_fill_gradient(low = "steel blue", high = "red") +
labs(
  title = "Debt-to-Income Ratio by State",
  x = "State",
  y = "Average Debt-to-Income Ratio",
  fill = "Ratio") + theme_minimal()
```



Interpretation:

This graphic demonstrates the differences in debt to income ratio by state.

Conclusion:

This analysis provided insight into the predictive power of tuition-related predictors and their influence on the debt-to-income ratio. The model created in this analysis was moderate at predicting low- to mid-sized values but performed poorly at predicting larger values. This model is a great exploratory segway to a more robust approach. This analysis demonstrated some of the limitations imposed by narrowing data to a scope such as the state level. This model demonstrates predictive power; with more significant predictors, it can be expanded to more accurately explain the factors contributing to differences in debt-to-income ratios across states.